

End-to-End Automation Framework for BlazeDemo Flight Booking Application

Problem Statement:

BlazeDemo is a publicly available demo website designed to mimic a real-world flight booking platform. Your task is to **design and implement a modular, scalable automation testing framework** using **Java, Selenium WebDriver, and TestNG** to validate all key functionalities.

This project simulates how automation frameworks are structured in companies and provides hands-on experience building them from scratch.

Objective:

- Understand the structure of a flight booking application.
- Create a reusable Selenium framework using Java and TestNG.
- Automate real user journeys.
- Handle positive and negative test cases.
- Use Maven to manage dependencies.
- Prepare for future integrations like CI/CD and advanced reporting.

Application Under Test:

URL: <https://blazedemo.com>

Key Pages

Page	Description
Home Page	Choose departure and destination cities.
Reserve Page	Lists available flights.
Purchase Page	Enter passenger and payment information.
Confirmation Page	Displays booking confirmation.

Step-by-Step Implementation Guide:

Step 1: Analyze the Application

Understand workflows: select cities → search → choose flight → purchase → confirm booking.

Step 2: Set Up Project

- Create a **Maven project** in your IDE (e.g., IntelliJ IDEA, Eclipse).
 - Add dependencies to pom.xml:
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Step 3: Design Framework (Page Object Model)

Packages

com.blazedemo.pages
com.blazedemo.tests
com.blazedemo.utils (optional)

Page Classes

Class	Actions Covered
HomePage.java	Select cities, click “Find Flights”.
ReservePage.java	Choose a flight. PurchasePage.java Enter passenger C payment info.
ConfirmationPage.java	Validate success message.

Step 4: Write a Basic Smoke Test

- Create FlightBookingTest.java under testpackage.
 - Automate booking a flight from Boston to New York.
 - Assert that the confirmation page is displayed.
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Step 5: Add Data-Driven Testing

- Use `@DataProvider` in TestNG to provide multiple user data sets (e.g., different names, addresses, card details).
 - Verify each booking is successful.
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Step 6: Add Negative Test Cases

Scenario	Expected Outcome
Submit blank credit card	Confirmation page should not appear.
Submit non-numeric credit card	Proper validation or error behavior.
Leave required fields blank	Proper validation or error behavior.
Same departure & destination city	Booking should not proceed.

Step 7: Organize Tests in Suite

- Create testng.xml.
- Add all test classes and define groups (e.g., smoke, functional, negative).

Step 8: Run & Debug

- Run via IDE or CLI (mvn test).
- Fix waits, locators, or timeouts as needed.
- Add assertions for validations.

Step 9: Optimize & Maintain

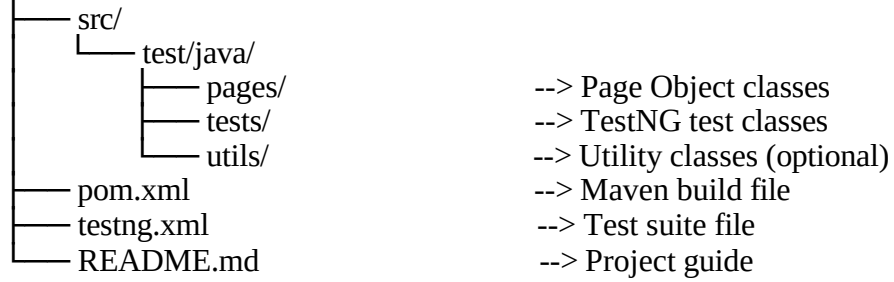
- Move driver setup to a Base class.
- Use @BeforeMethod, @AfterMethod.
- Add optional screenshot capture on failure.
- Keep code modular and reusable.

Test Scenarios to Automate

ID	Scenario Description	Type
TC01	Verify homepage loads and dropdowns visible	Smoke
TC02	Search flights with valid cities	Functional
TC03	Complete a flight booking	Functional
TC04	Multiple bookings with different data sets	Data-driven
TC05	Blank credit card	Negative
TC06	Invalid credit card characters	Negative
TC07	Same departure and destination city	Negative

Suggested Folder Structure

blazedemo-automation/



Expected Learning Outcomes:

- Design scalable Selenium frameworks using Page Object Model.
- Apply TestNG annotations, assertions, and data providers.
- Use Maven for dependency management.
- Write and organize positive & negative test cases.
- Gain practical skills to prepare for real-world automation tasks.

